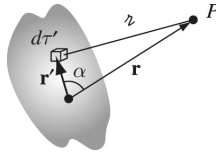


Tuesday Warm-up: Potentials

Prof. Jordan C. Hanson

September 30, 2025

$$\begin{aligned} P_0(x) &= 1 \\ P_1(x) &= x \\ P_2(x) &= (3x^2 - 1)/2 \\ P_3(x) &= (5x^3 - 3x)/2 \\ P_4(x) &= (35x^4 - 30x^2 + 3)/8 \\ P_5(x) &= (63x^5 - 70x^3 + 15x)/8 \end{aligned}$$



- The $\Theta(\theta)$ solutions are known as *Legendre polynomials*, $P_l(x)$, and $x = \cos \theta$, with the **Rodrigues** generating function

$$P_l(x) = \frac{1}{2^l l!} \left(\frac{d}{dx} \right)^l (x^2 - 1)^l \quad (3)$$

Derive $P_0(x)$, $P_1(x)$, and $P_2(x)$.

Figure 1: (Left) The first few Legendre polynomials. (Right) The angle α between $\mathbf{r}$ and $\mathbf{r}'$.

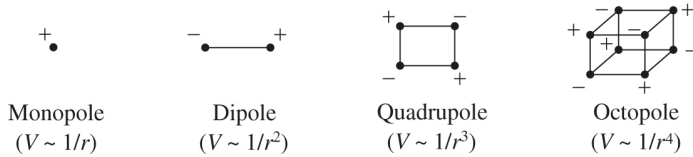


Figure 2: Configurations of point charges.

1 Memory Bank

- Laplace's equation** in spherical coordinates, with azimuthal symmetry:

$$\frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) + \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) = 0 \quad (1)$$

- Multipole expansion** in spherical coordinates:

$$V(r, \theta) = \frac{1}{4\pi\epsilon_0} \sum_{n=0}^{\infty} \frac{1}{r^{n+1}} \int (r')^n P_n(\cos \alpha) \rho(\mathbf{r}') d\tau' \quad (2)$$

2 Separation of Variables

- A hollow spherical shell of radius R has a potential $V_0(\theta) = V_0$ for $\theta \leq \pi/2$, and 0 for $\theta > \pi/2$. Find the voltage outside the shell.

- (a) Separate the solution for the voltage like $V(r, \theta) = R(r)\Theta(\theta)$. Isolate the $R(r)$ and r factors from the Θ and θ factors. (b) Set the ordinary differential equation for $R(r)$ equal to the constant $l(l+1)$, where l is an integer. Show that the solutions are linear combinations of r^l and $r^{-(l+1)}$.

- Write the solution for $V(r, \theta)$ as an infinite sum, and apply the boundary condition for $r \rightarrow \infty$.

- Apply the boundary condition at $r = R$, and use Fourier's trick to solve for the coefficients. You will need the following orthogonality condition:

$$\int_{-1}^1 P_l(x) P_l'(x) dx = \int_0^\pi P_l(\cos \theta) P_l'(\cos \theta) \sin \theta d\theta = \frac{2}{2l+1} \delta_{ll'} \quad (4)$$

3 Multipole Expansion

- Using the multipole expansion, generate the voltage of a physical (a) monopole, and (b) dipole. (c) What is the electric field of the dipole?